

Slides 4.14: Build the Model

Slide outline for Lesson 4.14. Day 1 has ten slides, Day 2 has nine, Day 3 has ten. Keep the conversion line on the board all three days: at 1/2 inch equals 1 foot, 1 foot is 0.5 inch, 3 feet is 1.5 inches, 8 feet is 4 inches. At 1 inch equals 1 foot, 3 feet is 3 inches and 8 feet is 8 inches. *[Sal: photograph every finished model on Day 3, no names and no faces. They are next year's hook and the website's project page.]*

Day 1

Slide 1: Do now

- Draw your client's room as a box with the walls folded down flat, a net
- The floor in the middle, four walls around it
- Label which wall has the door and which has the window
- Two minutes to draw, one to compare Notes: Day 9 of the floor plan stack. Anyone whose Lesson 4.13 exit card came in over budget fixes the Budget Sheet now instead, and builds after it balances.

Slide 2: Build the Model

- Lesson 4.14, Day 1 of 3
- Day 1: walls, floor, openings. Day 2: color, then furniture. Day 3: finish, labels, justification.
- A model is the plan you can walk around
- What does a model tell the client that a floor plan does not? Image: a finished shoebox bedroom model beside the floor plan of the same room. Notes: Hold up a box and a plan from a previous class, no name on it. "This is the same room. One you can read, one you can hold. By Friday yours is the one you can hold."

Slide 3: The model scale

- A shoebox is about 13 inches by 8 inches inside
- At 1/2 inch equals 1 foot that holds a room 26 feet by 16 feet, so every client room fits
- Bigger box or a foam board base: 1 inch equals 1 foot is allowed
- Write the scale on the rim of the box or the edge of the base Image: a shoebox with the lid off and "1/2 inch = 1 foot" written on the rim.

Slide 4: Mark the floor

- Count the squares on your plan, convert to feet, convert to model inches
- Draw the room's outline on the box floor with a ruler
- If the room is smaller than the box, the extra floor is outside the room
- Tiny house: 8 by 24 feet is 4 inches by 12 inches Image: a ruler on the floor of a shoebox with a pencil line drawn along it.

Slide 5: Doors and windows

- Find the wall and the spot from your plan, not from memory
- A 3-foot door is 1.5 inches at 1/2 inch equals 1 foot, or 3 inches at 1 inch equals 1 foot
- Bring the box to my cutting station for the door opening
- The window is cut at the station or drawn as a frame with a marker Image: a box wall with a door cut out and a window drawn in marker. Notes: Craft knives stay at my station. *[Sal: your room's rule on whether a supervised grade 8 student may cut.]*

Slide 6: Extra walls

- Closet: a card strip the closet's depth, tab folded at the bottom, glued in
- Tiny house bathroom: two card walls, 3 by 5 feet at scale, with a gap for the door
- Tiny house loft: a card shelf glued across one end, 4 feet (2 inches) below the rim
- A room is 8 feet high, which is 4 inches at our scale, so the box wall is the room wall Image: a card strip folded into an L and glued down as a closet wall.

Slide 7: The floor

- Cut paper, felt, fabric, or a printed wood or tile pattern to the room's floor size
- Glue it down
- A rug goes in later, as its own piece, only if the client allows one Image: felt glued into the floor of the box.

Slide 8: Craft day rules

- Sleeves up, hair back, same as a lab
- Glue guns: one station, one adult, low temperature, nobody walks with a hot gun
- A burn is a burn: cool water, tell me (Lesson 1.3)
- Tell me now if hot glue fumes or foam bother you. We have another way to do every step.
- Scraps go in the scrap bin, not the trash. Scraps are tomorrow's furniture. Notes: Say the allergy line out loud on Day 1 so nobody has to ask in front of the class. Keep the glue gun station away from the traffic path with a heat-safe mat under it.

Slide 9: Station check

- Glue caps on
- Scraps in the scrap bin
- Model on the shelf with your team name on masking tape
- Table wiped, floor clear Notes: Three minutes. Call it at the same time every day of this build.

Slide 10: Thumbs check

- Up: my walls and floor match my plan
- Sideways: one thing is off
- Down: I need a new box Notes: Fix the downs before Day 2. Pre-cut foam board bases are on the counter for anyone who brought no box.

Day 2

Slide 11: Do now

- Draw one wall of your room straight on, an elevation, at 1/2 inch equals 1 foot
- The floor line, the window, and the furniture that sits against that wall
- Rooms are usually 8 feet high
- Two minutes to draw, one to compare Notes: Day 10 of the stack. This is the first time students see the room in height, which is the point of today.

Slide 12: Color first, then furniture

- Lesson 4.14, Day 2
- Pull your Lesson 4.8 color card
- Main color on the walls, accent on the focal point
- You cannot paint behind a bed. Color goes on first. Image: three walls covered in pale blue paper and one wall in a deeper blue. Notes: A team with no color card picks one of the three schemes on the build guide: monochromatic, analogous, complementary. Name it; the label asks for it on Day 3.

Slide 13: Furniture from scrap, in ninety seconds

- Bed: a folded card box with a fabric scrap for the blanket
- Desk: a card rectangle on four toothpick legs, open underneath for a wheelchair user
- Dresser: a stacked card block with the drawers drawn on
- Lamp: a paper straw with a paper cone Image: those four pieces sitting next to a ruler. Notes: Build all three live, fast. Speed is the message: a model is scrap and twenty minutes, not a craft project.

Slide 14: To scale, every piece

- Use the cut-out sheet sizes from Handout 04.12, page 2
- At 1/2 inch equals 1 foot, a 3 by 6.5 foot twin bed is 1.5 by 3.25 inches
- A crib is 2.5 by 4.5 feet, so 1.25 by 2.25 inches
- Measure the model piece against the plan piece before you glue Notes: The common mistake is a bed the size of the room. Keep this slide up all period.

Slide 15: Build, 25 minutes

- Color the walls first
- Build every piece on your plan, needs first
- Place each piece where the plan puts it
- Do not glue a piece down until it matches the plan
- If a piece looks better somewhere else, change the plan too, in pencil, with a note Image: a card bed with a fabric blanket held next to the plan showing the bed on the same wall. Notes: Circulate and hold every team to the plan: your plan has the bed on the window wall and your model has it on the door wall. Which one is right?

Slide 16: While you build, I will ask

- What scale is your model? How long is your bed in inches here, and in feet in real life?
- Which wall has the focal point, and what color is on it?
- Is the room balanced, or is everything on one wall?
- Is that bed in proportion to the room? Notes: Lessons 4.6 and 4.7 vocabulary, used on a real object. Push for the word, not just the answer.

Slide 17: What the plan hid

- The plan said the desk fits under the window. Now the wall has height. Does the sill hit the desk?
- A tiny house team put the couch under the loft. Can a person stand up there?
- What does a model show that a plan cannot? Notes: These are the questions that make the model worth three days. Take two answers and let the class find more.

Slide 18: Station check

- Glue caps on, scraps in the bin
- Model on the shelf, team name on it Notes: Three minutes, same as yesterday.

Slide 19: Exit card

- My color scheme is ____
- The focal point is ____
- Tomorrow I still need to make ____ Notes: Sort the cards. The teams with a long “still need to make” list get the first ten minutes of Day 3 for furniture and finish labels during Lesson 4.15’s first minutes if they have to.

Day 3

Slide 20: Do now

- Draw a plan of any room with a wheelchair turning circle in it, a circle 5 feet across
- Where does it fit?
- What did you move to make room?
- Two minutes to draw, one to compare Notes: Day 11 of the stack, the last one before the stack is turned in with the project in Lesson 4.15.

Slide 21: Finish, label, justify

- Lesson 4.14, Day 3
- Glue the furniture, add the last pieces
- Three cards go on the model: GREEN, SAFE, WHY
- The client card goes on the outside. The scale goes on the rim. Your names go on the bottom. Image: a finished model with three label cards inside and the client card glued to the front wall.

Slide 22: The GREEN label

- One green choice from Lesson 4.5
- An LED lamp, a secondhand piece, low-VOC paint, a recycled-fiber rug, a window placed for daylight, a plant
- Say why it matters for this client, not just in general
- Frame: “This is . ***It is green because the client*** .” Notes: “It is blue like the ocean” is not a green choice. Push until the sentence names energy, water, reuse, waste, or cleaner air.

Slide 23: The SAFE label

- One safety feature from Lesson 4.2
- A night light on the path, cordless blinds, a smoke alarm on the ceiling, a rug taped down or no rug, a crib 36 inches from the window, a 36-inch clear path
- Say what it prevents: a fall, a trip, a burn, a cord hazard, a blocked exit
- Frame: “This is . ***It is safe because the client*** .” Notes: A smoke alarm on the ceiling is rare and correct. Praise it out loud when you see it.

Slide 24: The WHY card

- Two sentences
- One: the biggest design choice in this room, and the client need it serves
- Two: the principle of design it uses, and why
- Principles: balance, focal point, rhythm, proportion, emphasis, unity Image: a filled WHY card for the Okafor room. Notes: Model it on the board for Okafor. “The biggest choice is the bed on the long wall with 36 inches on the open side, because Grandpa gets in and out with a walker. It uses proportion, because a twin bed instead of a full leaves the floor open.”

Slide 25: Grade 8: the ACCESSIBLE label

- A fourth card for any universal design feature
- Bed height, desk knee space, closet rod height, switch height
- Write the measurement on the card
- Grade 8 stretch: build one wall elevation on card that lifts off to show the wall’s height and window Notes: Tiny house teams at grade 8 build the loft with the ladder to scale.

Slide 26: Finish, 22 minutes

- Glue every piece from the plan
- Add the lamp, the rug, the window covering, the plant, the hooks
- Fix anything out of scale
- Mount GREEN, SAFE, WHY, and the client card
- Write the scale on the rim and your names on the bottom Notes: Clipboard check, one line per team: matches the plan; color scheme named and applied; GREEN and SAFE each give a reason for this client; scale written; WHY has a need and a principle. Four or five yeses is a Meets on criterion 4 before the pitch.

Slide 27: No box, no problem

- The isometric grid page is the same grade
- The floor as a parallelogram at 1/2 inch equals 1 foot, walls up 8 feet
- Every piece of furniture drawn as a box to scale
- The color scheme colored in, GREEN and SAFE drawn as callouts with arrows
- A 3-D room in a free app the school allows also counts, printed or on a screen [*Sal: name the app*] Notes: Say this on all three days so nobody thinks the drawing is the consolation prize. Criterion 4 is scored the same way.

Slide 28: Self-check against the rubric

- Read Rubric 04, criterion 4
- Circle your level
- Fix one thing Notes: Five minutes. Hand Rubric 04 back for this. Tomorrow: the pitch and the gallery walk. The plan, the Budget Sheet, the model, and the whole do-now stack are turned in.

Slide 29: Sticky note out

- One sentence on a note, stuck on your model
- “The client will notice ___ first.” Notes: Read three aloud. Then the station check: glue caps on, scraps in the bin, models on the shelf by period.